

Cave: Core Formal System

A proof-style formalization of subjective trajectories

Prepared from the Cave framework notes

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Abstract

This document condenses the Cave framework into a sequence of definitions, update equations, invariants, and proof-style propositions. It treats a subjective trajectory not as a metaphysical claim about consciousness, but as a serialized subject-side state history produced when an external episode passes through a configured subject profile.

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1 External Episode

Let an external episode be a temporally ordered sequence of available world-side objects:

$$X_{1:T} = (O_1, O_2, \dots, O_T). \quad (1)$$

At each timestep, the active object set is

$$O_t = \{o_i : t \in I_i\}, \quad (2)$$

where each object has the form

$$o_i = (I_i, \phi_i, m_i, s_i, \ell_i, \mu_i). \quad (3)$$

Here

$$I_i = [t_i^{on}, t_i^{off}), \quad (4)$$

$$\phi_i = \text{feature description}, \quad (5)$$

$$m_i = \text{modality}, \quad (6)$$

$$s_i = \text{salience}, \quad (7)$$

$$\ell_i = \text{learning weight}, \quad (8)$$

$$\mu_i = \text{metadata}. \quad (9)$$

The object is *available* if it belongs to O_t , but availability is not yet subject-side uptake.

2 Subject Profile

A subject profile is a fixed configuration:

$$\theta = (\mathcal{S}, \alpha, \psi, \kappa, F_M, F_E, F_V, F_J, \rho, \pi, S_0), \quad (10)$$

where

$$\mathcal{S} = \text{sensorium}, \quad (11)$$

$$\alpha_t = \text{attention distribution}, \quad (12)$$

$$\psi(c_t) = \text{attention capacity effect}, \quad (13)$$

$$\kappa = \text{workspace compressor}, \quad (14)$$

$$F_M = \text{memory update rule}, \quad (15)$$

$$F_E = \text{expectation rule}, \quad (16)$$

$$F_V = \text{value evaluator}, \quad (17)$$

$$F_J = \text{objective evaluator}, \quad (18)$$

$$\rho_t = \text{exposure policy}, \quad (19)$$

$$\pi = \text{projection into inspection space}, \quad (20)$$

$$S_0 = \text{initial subject state}. \quad (21)$$

A trajectory is therefore not a property of the external episode alone:

$$\tau \neq \tau(X_{1:T}), \quad (22)$$

but rather

$$\tau = \tau(\theta, X_{1:T}). \quad (23)$$

3 Subject-Side State

At each time t , the model stores a full subject-side state:

$$S_t = (t, O_t, O'_t, A_t, U_t, W_t, E_t, P_t, \sigma_t, \eta_t, M_t, V_t, J_t, a_t, L_t). \quad (24)$$

The components are:

$$O_t = \text{world-side active objects}, \quad (25)$$

$$O'_t = \text{exposed or sensed material}, \quad (26)$$

$$A_t = \text{attention state}, \quad (27)$$

$$U_t = \text{attended subject-side input}, \quad (28)$$

$$W_t = \text{workspace content}, \quad (29)$$

$$E_t = \text{expectation}, \quad (30)$$

$$P_t = \text{prediction error}, \quad (31)$$

$$\sigma_t = \text{surprise}, \quad (32)$$

$$\eta_t = \text{learning rate}, \quad (33)$$

$$M_t = \text{memory}, \quad (34)$$

$$V_t = \text{value state}, \quad (35)$$

$$J_t = \text{objective state}, \quad (36)$$

$$a_t = \text{action or exposure modulation}, \quad (37)$$

$$L_t = \text{topology-like accumulated field}. \quad (38)$$

The state update is

$$S_{t+1} = F_t(S_t, O_t; \theta), \quad (39)$$

and the subjective trajectory is the ordered state history:

$$\tau(\theta, X_{1:T}) = (S_0, S_1, \dots, S_T). \quad (40)$$

4 Exposure and Sensing

External availability is filtered by exposure:

$$O'_t = \rho_t(O_t, S_t). \quad (41)$$

Then sensing applies the subject's sensorium:

$$x_t = \mathcal{S}_\theta(O'_t). \quad (42)$$

If no sensor exists for modality m , then

$$\mathcal{S}_{\theta,m}(O'_t) = 0. \quad (43)$$

Therefore

$$o_i \in O_t \not\Rightarrow o_i \in x_t. \quad (44)$$

World-side presence does not imply subject-side presence.

5 Attention Gate

Attention consists of capacity and allocation:

$$A_t = (c_t, \alpha_t), \quad (45)$$

with

$$\sum_i \alpha_{t,i} = 1 \quad (46)$$

when $c_t > 0$.

The attention gate is

$$g_t = \psi(c_t)\alpha_t. \quad (47)$$

For external modality m :

$$\tilde{x}_{t,m} = \psi(c_t)\alpha_{t,m}x_{t,m}. \quad (48)$$

For internal expectation:

$$\tilde{E}_t = \psi(c_t)\alpha_{t,\text{exp}}E_t. \quad (49)$$

Thus attention changes gain, not feature coordinates:

$$\phi(o_i) \mapsto \phi(o_i), \quad (50)$$

but its effective contribution becomes

$$\psi(c_t)\alpha_{t,m}\phi(o_i). \quad (51)$$

If $c_t = 0$, then

$$\psi(c_t) = 0, \quad (52)$$

and therefore

$$U_t = 0, \quad P_t = 0, \quad \eta_t = 0, \quad \Delta M_t = 0, \quad D_t = 0. \quad (53)$$

6 Attended Input

The attended subject-side input is

$$U_t = \psi(c_t)\alpha_t \odot \mathcal{S}_\theta(O'_t), \quad (54)$$

or compactly

$$U_t = G_t x_t, \quad (55)$$

where

$$G_t = \psi(c_t) \text{diag}(\alpha_t). \quad (56)$$

7 Expectation

Expectation must be computed from prior memory, not current-updated memory:

$$E_t = F_E(M_{t-1}, S_{t-1}; \theta). \quad (57)$$

This imposes the temporal order:

$$M_{t-1} \longrightarrow E_t \longrightarrow U_t \longrightarrow P_t \longrightarrow M_t. \quad (58)$$

The model must not compute

$$E_t = F_E(M_t), \quad (59)$$

because that would allow the present input to contaminate the prediction it is supposed to test.

8 Prediction Error and Surprise

Prediction error compares admitted actual input with admitted expectation:

$$P_t = U_t - \tilde{E}_t. \quad (60)$$

Equivalently,

$$P_t = \psi(c_t) (\alpha_{t,\text{actual}} \odot x_t - \alpha_{t,\text{exp}} \odot E_t). \quad (61)$$

Surprise is a scalar or structured function of error:

$$\sigma_t = f_\sigma(P_t). \quad (62)$$

For example:

$$\sigma_t = \|P_t\|. \quad (63)$$

Learning rate is derived from surprise, value, salience, or profile-specific rules:

$$\eta_t = f_\eta(\sigma_t, V_t, \theta). \quad (64)$$

9 Vector Memory

Vector memory updates from subject-side input, not external input:

$$M_{t+1} = F_M(M_t, U_t, P_t, \eta_t; \theta). \quad (65)$$

A simple additive form is:

$$M_{t+1} = M_t + \eta_t P_t. \quad (66)$$

Alternatively, if memory incorporates attended input directly:

$$M_{t+1} = (1 - \eta_t) M_t + \eta_t U_t. \quad (67)$$

The key invariant is

$$\Delta M_t \text{ depends on } U_t, \text{ not directly on } O_t. \quad (68)$$

So:

$$O_t \neq \emptyset \quad \text{and} \quad U_t = 0 \quad \Rightarrow \quad \Delta M_t = 0. \quad (69)$$

10 Object Memory

For an object o active over

$$I_o = [t_o^{on}, t_o^{off}), \quad (70)$$

its retained trace strength is

$$r_o^0 = \sum_{t=t_o^{on}}^{t_o^{off}} \psi(c_t) \alpha_t(o) q_o(t), \quad (71)$$

where $q_o(t)$ is the transduced object contribution.

After closure, the retained trace decays:

$$r_o(\tau) = r_o^0 D(\tau), \quad (72)$$

with

$$D'(\tau) \leq 0. \quad (73)$$

Thus object memory is not a list of all objects that existed. It is a list of objects that affected the subject sufficiently to leave a trace.

11 Workspace Compression

The workspace receives attended input and compresses it:

$$W_t = \kappa_\theta(U_t; K_t), \quad (74)$$

where K_t is workspace capacity.

Compression loss may be recorded as

$$C_t = d(U_t, W_t), \quad (75)$$

or

$$C_t = \|U_t - \kappa_\theta^{-1}(W_t)\|. \quad (76)$$

The workspace invariant is

$$U_t \neq W_t \quad (77)$$

in general, because what is sensed and attended may exceed what can be carried forward.

12 Value and Objective State

Prediction error is distinct from value.

Valence is computed as

$$V_t = F_V(U_t, O'_t, M_t, \theta). \quad (78)$$

Objective state is computed from subject-side quantities:

$$J_t = F_J(P_t, \sigma_t, C_t, V_t, \eta_t, A_t, M_t; \theta). \quad (79)$$

A generic scalar objective might be:

$$J_t = w_\sigma \sigma_t + w_C C_t + w_V V_t + w_A \text{Cost}(A_t) + w_L \eta_t. \quad (80)$$

The separation principle is

$$P_t \neq V_t. \quad (81)$$

A stimulus can be expected and painful, or surprising and pleasant. Mismatch and desirability must remain separate coordinates.

13 Exposure Policy

Future exposure is modulated by current state:

$$\rho_{t+1} = F_\rho(\rho_t, J_t, V_t, \sigma_t, M_t; \theta). \quad (82)$$

Then later sensing depends on prior valuation and exposure:

$$O'_{t+1} = \rho_{t+1}(O_{t+1}, S_{t+1}). \quad (83)$$

This closes the minimal loop:

$$V_t, J_t \longrightarrow \rho_{t+1} \longrightarrow O'_{t+1} \longrightarrow U_{t+1} \longrightarrow S_{t+2}. \quad (84)$$

14 Topology-Like State

Let

$$\pi : \mathcal{F} \rightarrow \mathcal{Z} \quad (85)$$

be a projection from subject-side feature space into an inspection space.

The topology-like accumulated field is

$$L_{t+1}(z) = \lambda L_t(z) + D_t(z), \quad (86)$$

where

$$0 \leq \lambda \leq 1 \quad (87)$$

and $D_t(z)$ is the current deposit.

A generic deposit may combine attended input, expectation, memory, and correction:

$$D_t(z) = d_U(z; \pi(U_t)) + d_E(z; \pi(E_t)) + d_M(z; \pi(M_t)) + d_P(z; \pi(P_t)). \quad (88)$$

The topology invariant is:

$$L_t \text{ is a projection-derived accumulated diagnostic, not the trajectory itself.} \quad (89)$$

15 Episode Contract

An Episode serializes:

$$\mathcal{E} = (\text{source}, \mathcal{V}, \text{EpisodeInput}_{1:n}, \text{EpisodeObservation}_{1:T}, T, \mu). \quad (90)$$

Each observation records:

$$\text{Obs}_t = (t, O_t, O'_t, E_t, U_t, M_t, P_t, \sigma_t, \eta_t, A_t, V_t, J_t, L_t, \mu_t). \quad (91)$$

The Episode is not a visualization. It is the serialized trace of the subject-side computation.

16 Views as Projections

A frame is selected from the trajectory:

$$F_t = \text{Frame}(S_t). \quad (92)$$

A view is a projection:

$$v_i : F_t \rightarrow R_{i,t}. \quad (93)$$

Pixels are produced only afterward:

$$\text{Episode} \longrightarrow \text{EpisodeFrame} \longrightarrow \text{ViewState} \longrightarrow \text{pixels}. \quad (94)$$

The no-backflow rule is

$$\frac{\partial S_t}{\partial R_{i,t}} = 0, \quad (95)$$

or conceptually:

$$R_{i,t} \not\rightarrow S_t. \quad (96)$$

A rendering may reveal, compress, color, arrange, or animate the trajectory, but it may not alter attention, memory, expectation, value, or error.

17 Core Propositions

Proposition 1 (Same Episode, Different Profiles, Different Trajectories). *Given the same external episode $X_{1:T}$, two profiles θ_a and θ_b may produce different trajectories:*

$$\theta_a \neq \theta_b \Rightarrow \tau(\theta_a, X_{1:T}) \neq \tau(\theta_b, X_{1:T}). \quad (97)$$

Proof sketch. The transition rule depends on θ :

$$S_{t+1} = F_t(S_t, O_t; \theta). \quad (98)$$

If θ_a and θ_b differ in sensorium, attention, memory, value, or exposure, then for the same O_t :

$$U_t^a = \psi(c_t^a) \alpha_t^a \odot \mathcal{S}_{\theta_a}(O_t') \quad (99)$$

may differ from

$$U_t^b = \psi(c_t^b) \alpha_t^b \odot \mathcal{S}_{\theta_b}(O_t'). \quad (100)$$

Therefore P_t , M_t , V_t , J_t , and L_t may also differ. Hence the full state histories may differ. $\square$

Proposition 2 (Availability Does Not Imply Uptake). *If an object is active externally, it does not necessarily affect the subject:*

$$o_i \in O_t \not\Rightarrow \Delta M_t \neq 0. \quad (101)$$

Proof sketch. Memory changes only through attended subject-side input:

$$M_{t+1} = F_M(M_t, U_t, P_t, \eta_t). \quad (102)$$

But

$$U_t = \psi(c_t) \alpha_t \odot \mathcal{S}_{\theta}(O_t'). \quad (103)$$

If the subject lacks the relevant sensor, $\mathcal{S}_{\theta}(o_i) = 0$, or if attention collapses, $\psi(c_t) = 0$, then $U_t = 0$ and therefore $\Delta M_t = 0$. $\square$

Proposition 3 (Expectation Must Precede Current Update). *A valid expectation at time t must be computed from prior memory:*

$$E_t = F_E(M_{t-1}), \quad (104)$$

not

$$E_t = F_E(M_t). \quad (105)$$

Proof sketch. Prediction error is defined as $P_t = U_t - E_t$. If E_t were computed from M_t , and M_t already incorporated U_t , then the expected state would partially contain the actual present input. This would suppress or distort error. Therefore valid temporal ordering requires

$$M_{t-1} \rightarrow E_t \rightarrow P_t \rightarrow M_t. \quad (106)$$

$\square$

Proposition 4 (Attention Changes Consequence, Not Feature Identity). *For an object o with feature representation $\phi(o)$, attention scales contribution but does not rewrite the representation:*

$$\tilde{\phi}_t(o) = \psi(c_t) \alpha_t(o) \phi(o). \quad (107)$$

The feature address remains $\phi(o)$.

Proof sketch. The gate is multiplicative:

$$U_t = \psi(c_t)\alpha_t \odot \mathcal{S}_\theta(O'_t). \quad (108)$$

This changes the magnitude of contribution to the state update. It does not apply a transformation $\phi(o) \mapsto \phi'(o)$. Therefore attention affects causal force in the trajectory, not the object's underlying feature geometry. $\square$

Proposition 5 (Prediction Error Is Not Value). *Prediction error and value are distinct coordinates:*

$$P_t \neq V_t. \quad (109)$$

Proof sketch. Prediction error is computed by comparison:

$$P_t = U_t - E_t. \quad (110)$$

Value is computed by evaluation:

$$V_t = F_V(U_t, O'_t, M_t, \theta). \quad (111)$$

A painful expected stimulus can have $P_t \approx 0$ but $V_t \ll 0$. A pleasant surprise can have $\|P_t\| \gg 0$ and $V_t > 0$. Thus mismatch and desirability are formally independent. $\square$

Proposition 6 (Views Are Observationally Inert). *For any view v_i ,*

$$v_i : F_t \rightarrow R_{i,t}, \quad (112)$$

the rendered result cannot alter the trajectory:

$$R_{i,t} \nrightarrow S_t. \quad (113)$$

Proof sketch. The rendering pipeline is one-way:

$$\text{Episode} \rightarrow \text{EpisodeFrame} \rightarrow \text{ViewState} \rightarrow \text{pixels}. \quad (114)$$

Since S_t is computed before F_t , and $R_{i,t}$ is computed after F_t , there is no permitted update rule of the form $S_{t+1} = G(S_t, R_{i,t})$. Therefore changing the renderer may change pixels but not the modeled subject-side state. $\square$

Proposition 7 (A Subjective Trajectory Is Not a Replay of Inputs). *The external sequence is $X_{1:T}$. The subjective trajectory is*

$$\tau(\theta, X_{1:T}) = (S_0, \dots, S_T). \quad (115)$$

Since each S_t depends on prior state, attention, memory, expectation, value, and exposure,

$$S_{t+1} = F_t(S_t, O_t; \theta), \quad (116)$$

the trajectory contains relations absent from the raw input stream.

Proof sketch. The raw input stream can list what occurred. The trajectory also preserves the chain

$$\text{sensed} \rightarrow \text{retained} \rightarrow \text{expected} \rightarrow \text{actual} \rightarrow \text{error} \rightarrow \text{learning} \rightarrow \text{future exposure}. \quad (117)$$

Therefore $\tau(\theta, X_{1:T}) \neq X_{1:T}$. $\square$

18 Minimal Axioms

Axiom 1 (Availability is not uptake).

$$o \in O_t \not\Rightarrow o \in U_t. \quad (118)$$

Axiom 2 (Uptake is subject-dependent).

$$U_t = \psi(c_t)\alpha_t \odot \mathcal{S}_\theta(\rho_t(O_t)). \quad (119)$$

Axiom 3 (Expectation is prior-state prediction).

$$E_t = F_E(M_{t-1}, S_{t-1}; \theta). \quad (120)$$

Axiom 4 (Error compares admitted actual with admitted expected).

$$P_t = U_t - \tilde{E}_t. \quad (121)$$

Axiom 5 (Memory updates from subject-side signal).

$$M_{t+1} = F_M(M_t, U_t, P_t, \eta_t; \theta). \quad (122)$$

Axiom 6 (Value is distinct from prediction error).

$$V_t = F_V(U_t, O'_t, M_t; \theta), \quad V_t \neq P_t. \quad (123)$$

Axiom 7 (Future exposure depends on current state).

$$\rho_{t+1} = F_\rho(\rho_t, J_t, V_t, \sigma_t, M_t; \theta). \quad (124)$$

Axiom 8 (Views are projections, not model updates).

$$Episode \rightarrow Frame \rightarrow View \rightarrow Pixels, \quad Pixels \not\rightarrow S_t. \quad (125)$$

19 One-Line Formal Summary

The Cave framework can be expressed as:

$$\boxed{\tau(\theta, X_{1:T}) = (S_0, F_1(S_0, O_1; \theta), \dots, F_T(S_{T-1}, O_T; \theta))} \quad (126)$$

where each update decomposes as:

$$\boxed{\begin{aligned} &O_t \rightarrow \rho_t(O_t) \rightarrow \mathcal{S}_\theta(\rho_t(O_t)) \rightarrow U_t \rightarrow W_t \\ &\rightarrow (E_t, P_t, \sigma_t, \eta_t) \rightarrow (M_{t+1}, V_t, J_t, \rho_{t+1}, L_{t+1}) \end{aligned}} \quad (127)$$

The central claim is:

$$\boxed{\text{Experience-like structure is not the input stream, but the subject-dependent state path produced from it.}}$$